MATH 223: Linear Algebra

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Abstract

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1 Introduction

2 Prerequisite Knowledge

2.1 Introduction to Proof

2.1.1 Direct Proof

The basic idea of a direct proof of $P \implies Q$ is:

1. Assume P is true,
2. Use P to show that Q is true (the hard part).

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a, b are consecutive integers, meaning $b = a + 1$. So

$$a + b = a + (a + 1) = 2a + 1.$$

Thus b is odd if and only if a is odd. □

2.1.2 Proof by Contradiction

The idea is that a proposition is either true or false, but not both. We get a contradiction when we can show a statement is both true and false, showing our initial assumptions are inconsistent. We use this to show $P \implies Q$ by assuming both P and not Q are simultaneously true and deriving a contradiction.

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a and b are consecutive integers. Assume that $(a + b)$ is not odd. If $(a + b)$ is not odd, then no integer k such that $(a + b) = 2k + 1$. But $a + b = a + (a + 1) = 2a + 1$. We have shown that $(a + b) \neq 2k + 1$ but $(a + b) = 2a + 1$, a contradiction. □

2.1.3 Proof by Contrapositive

The idea is to prove $P \implies Q$ by instead proving not $Q \implies$ not P.

Example. *If n^2 is even, then n is even.*

Proof. We show the contrapositive: if n is odd, then n^2 is odd. Assume n is odd, $\exists k_1 \in \mathbb{Z}, n = 2k_1 + 1$. So

$$n^2 = (2k_1 + 1)^2 = 4k_1^2 + 4k_1 + 1 = 2(2k_1^2 + 2k_1) + 1.$$

Let $k_2 = 2k_1^2 + 2k_1$. Then $n^2 = 2k_2 + 1$, so n^2 is odd. □

2.2 Notation

2.2.1 Sets

Sets are a grouping of objects.

Set	Meaning	Examples
$\mathbb{N}$	The set of natural numbers	(0, 1, 2, 3, ...)
$\mathbb{Z}$	The set of integers	(..., -3, -2, -1, 0, 1, 2, 3, ...)
$\mathbb{Q}$	The set of rational numbers	$\mathbb{Q} = \frac{a}{b} \mid \forall a, b \in \mathbb{Z}\ and\ b \neq 0$
$\mathbb{R}$	The set of all rational and all irrational numbers	$(..., -1, 0, \frac{1}{4}, 1, 1000, ...)$
$\mathbb{C}$	The set of all complex numbers	$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\ and\ i \subseteq \sqrt{-1}\}$.

We have the following relationships between sets:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

2.2.2 Symbols

Symbol	Meaning
$\subseteq$	is a subset of or equal to
$\subset$	is a strict subset of
$\in$	is an element of
$\forall$	for all
$\exists$	there exists
$\emptyset$	empty set
$\Rightarrow$	implies
$\Leftrightarrow$	if and only if
$\cong$	is isomorphic to[1]

2.3 Complex Numbers

2.3.1 Definition and the Fundamental Theorem of Algebra

Definition 1 (Complex Number)**.** *A complex number is of the form: $z = x + iy$ where $x, y \in \mathbb{R}$ and i is the imaginary unit $i = \sqrt{-1}$.*

Theorem 1 (Fundamental Theorem of Algebra)**.** *The FTA states that any non-constant, single-variable polynomial[2] with complex coefficients has at least one root in $\mathbb{C}$.*

Remark 1. *If we have a polynomial f of degree n, then it has n roots, where each root can have a multiplicity[3].*

[1] is isomorphic to = structurally the same as.

[2] A polynomial is a function such as $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ where $a_i \in \mathbb{R}$ or $\mathbb{C}$ and $n \in \mathbb{N}$.

[3] The multiplicity of a root represents how many times the root occurs in the polynomial.

Example. $(x-1)^2$ *has degree 2 but only one root, which is 1, with multiplicity 2.*

We can factorize a polynomial $f = a_n x^n + \cdots + a_1 x + a_0$ as $f = a(x-z_1)(x-z_2)\cdots(x-z_n)$, where z_i are the roots of f in $\mathbb{C}$.

2.3.2 Arithmetic Operations

We define the following operations on complex numbers $z = x + iy$ and $z' = x' + iy'$:

- Addition: $z + z' = (x + x') + i(y + y')$.
- Multiplication: $zz' = (xx' - yy') + i(xy' + yx')$.
- Inverse: $\dfrac{1}{z} = \dfrac{\overline{z}}{z\overline{z}} = \dfrac{x - iy}{x^2 + y^2} = \dfrac{x}{x^2 + y^2} + i\dfrac{-y}{x^2 + y^2}$.

Multiplying by a complex number z corresponds geometrically to:

$$\begin{cases} \text{a rotation by angle } \theta, \\ \text{a rescaling by factor } |z|. \end{cases}$$

2.3.3 Complex Conjugate

Definition 2 (Complex Conjugate)**.** *For $z = x + iy$, the complex conjugate is $\overline{z} = x - iy$.*

Basic properties:

- $\overline{\overline{z}} = z$
- $\overline{z + z'} = \overline{z} + \overline{z'}$
- $\overline{z \cdot z'} = \overline{z} \cdot \overline{z'}$

2.3.4 Geometric Interpretation and Modulus

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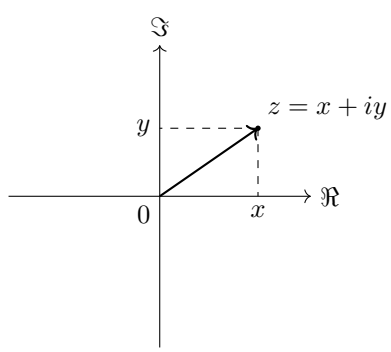
Definition 3 (Geometric interpretation)**.** *Every complex number $z = x + iy$ is identified with a point (x, y) in the complex plane:*

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}.$$



Definition 4 (Modulus)**.** *The modulus of z is*

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

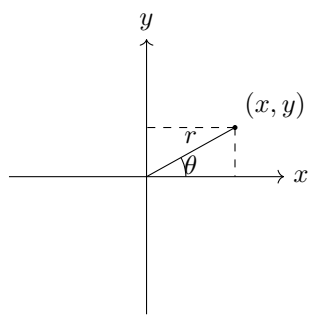
Geometrically, $|z|$ is the distance from the origin to (x, y).

The unit circle in the complex plane can be written as:

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} = \{z \in \mathbb{C} : |z| = 1\}.$$

2.3.5 Polar Coordinates and Polar Form

Definition 5 (Polar coordinates)**.** *Instead of (x, y), a point can be described by polar coordinates (r, θ), where r is the distance to the origin and θ is the angle with the positive x-axis.*



So $x = r\cos\theta$ and $y = r\sin\theta$, with $r = |z|$.

Definition 6 (Euler's formula)**.**

$$e^{i\theta} = \cos\theta + i\sin\theta. \tag{1}$$

Definition 7 (Polar and exponential form)**.** *If $z = x + iy$ with $r = \sqrt{x^2 + y^2}$, then $x = r\cos\theta$, $y = r\sin\theta$, so*

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}.$$

Definition 8 (Multiplication in polar form)**.**

$$z = re^{i\theta}, \quad z' = r'e^{i\theta'} \implies zz' = rr'e^{i(\theta+\theta')}.$$

Example.

$$(1 + i)^{32} = (\sqrt{2}\, e^{i\pi/4})^{32} = 2^{16}\, e^{i8\pi} = 2^{16}(\cos 8\pi + i\sin 8\pi) = 2^{16}.$$

2.3.6 Roots in $\mathbb{C}$

Definition 9 (n^{th} roots)**.** *For $z \in \mathbb{C}$, an nth root of z is $w \in \mathbb{C}$ with $w^n = z$.*

Example. *If $z = re^{i\theta}$, the solutions of $w^n = z$ are*

$$w_k = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, \ldots, n-1.$$

They are equally spaced on a circle of radius $r^{1/n}$ centered at the origin.

Definition 10 (Roots of unity)**.** *The nth roots of unity are solutions of $w^n = 1$:*

$$w_k = e^{i2\pi k/n}, \quad k = 0, 1, \ldots, n-1.$$

They lie on the unit circle and are equally spaced.

2.3.7 Hermitian Matrices

Definition 11 (Hermitian Matrix)**.** *A matrix $A \in M_n(\mathbb{C})$ is Hermitian if $A = A^*$, where $A^* = \overline{A}^T$. Equivalently,*

$$a_{ij} = \overline{a_{ji}} \quad \text{for all } i, j.$$

Example. $A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$ *is Hermitian since $A^* = A$.*

Remark 2. *All eigenvalues of a Hermitian matrix are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Problem 1 (Hermitian Matrix)**.** *Let $A, B \in M_n(\mathbb{C})$. Prove that $(AB)^* = B^* A^*$.*

2.4 Basic Algebraic Structures

2.4.1 Sets with Multiplication

Definition 12 (Set with multiplication)**.** *A set M is a set with multiplication if for any $a, b \in M$, the product $ab \in M$.*

Example. *$M_2(\mathbb{C})$ (all 2×2 complex matrices) and $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ are sets with multiplication.*

2.4.2 Invertibility

Definition 13 (Invertibility)**.** *Let $A \in M_n(k)$. If there exists $B \in M_n(k)$ with $AB = I_n$ or $BA = I_n$, then A is invertible and $B = A^{-1}$. Here I_n is the $n \times n$ identity matrix*[4]*.*

Remark 3. *If A is invertible, A^{-1} is unique*[5]*.*

[4] The identity matrix has 1s on its diagonal and 0s elsewhere.
[5] Unique: there is exactly one such element.

Example. $2 \in \mathbb{Z}$ *is* not *invertible since* $\frac{1}{2} \notin \mathbb{Z}$*, but* $2 \in \mathbb{R}$ *is invertible since* $\frac{1}{2} \in \mathbb{R}$*.*

Example. $1 + i \in \mathbb{C}$ *is invertible:* $\dfrac{1}{1+i} = \dfrac{1-i}{2} \in \mathbb{C}$ *(using the formula from* (2.3.2)*).*

Problem 2 (Invertibility)**.** *Show that if an inverse of* A *in* M *exists, then it is unique.*

Problem 3 (Invertibility 2)**.** *Let* K *be a field. Prove that* $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \in M_2(K)$ *is not invertible.*

2.4.3 Rings

Definition 14 (Ring)**.** *A ring is a set* R *with addition and multiplication satisfying:*

1. $(R, +)$ *is a commutative*[6] *and associative*[7] *group with zero and additive inverses.*
2. *Multiplication is associative with identity* 1*.*
3. *Distributivity:* $a(b + c) = ab + ac$ *and* $(a + b)c = ac + bc$*.*

Example. $(\mathbb{Z}, +, \cdot)$ *is a ring: it is closed under addition and multiplication, has zero element and additive inverses, multiplication is associative with identity 1, and multiplication distributes over addition.*

2.4.4 Fields

Definition 15 (Field)**.** *A field is a ring* (14) *in which every nonzero element has a multiplicative inverse.*

Example. $(\mathbb{Q}, +, \cdot)$ *is a field: it is a ring, and every nonzero* $q \in \mathbb{Q}$ *has* $q^{-1} \in \mathbb{Q}$ *with* $q \cdot q^{-1} = 1$*.*

Remark 4. $\mathbb{Z}$ *is a ring but not a field: for example, there is no integer* x *with* $2x = 1$*.*

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3 Vector Spaces

Definition 16 (Vector Space)**.** *Let* k *be a field. A vector space over* k *is a set* V *with addition and scalar multiplication satisfying:*

1. *Closure under addition:* $v_1 + v_2 \in V$

[6] $a + b = b + a$.
[7] $a + (b + c) = (a + b) + c$.

2. *Commutativity:* $v_1 + v_2 = v_2 + v_1$
3. *Associativity:* $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$
4. *Zero vector:* $\exists\, 0 \in V$ *with* $v + 0 = v$
5. *Additive inverses:* $\forall v \in V$, $\exists\, -v \in V$ *with* $v + (-v) = 0$
6. *Closure under scalar multiplication:* $\lambda v \in V$
7. *Compatibility:* $\lambda(\mu v) = (\lambda\mu)v$
8. *Identity:* $1 \cdot v = v$
9. *Distributivity over vectors:* $\lambda(v_1 + v_2) = \lambda v_1 + \lambda v_2$
10. *Distributivity over scalars:* $(\lambda + \mu)v = \lambda v + \mu v$

Remark 5. *We often say* abstract *vector space to emphasize that elements of V can be any objects (functions, tensors, matrices, etc.) satisfying these axioms — not just arrows or tuples in $\mathbb{R}^n$.*

3.1 Vectors and Linear Combinations

Vector operations in $\mathbb{R}^n$ are defined component-wise:

- Addition: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$.
- Scalar multiplication: $\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$, where $\lambda \in \mathbb{R}$.

Definition 17 (Linear combination)**.** *A linear combination of $v_1, \ldots, v_n$ is $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ for scalars $\lambda_i \in k$.*

Example (Computing a linear combination)**.** *Let $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$.*
Express $v = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$ as a linear combination of v_1 and v_2.
We need $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $\lambda_1 v_1 + \lambda_2 v_2 = v$, i.e.,

$$\lambda_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} \implies \begin{cases} \lambda_1 + 3\lambda_2 = 7 \\ 2\lambda_1 - \lambda_2 = 3. \end{cases}$$

From the second equation $\lambda_2 = 2\lambda_1 - 3$. Substituting: $\lambda_1 + 3(2\lambda_1 - 3) = 7 \implies 7\lambda_1 = 16 \implies \lambda_1 = \frac{16}{7}$, $\lambda_2 = \frac{11}{7}$. Hence

$$v = \tfrac{16}{7} v_1 + \tfrac{11}{7} v_2.$$

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3.2 Span and Standard Basis

3.2.1 Span

Definition 18 (Span)**.** *Let $A \subset \mathbb{R}^n$. The span of A is the set of all linear combinations of elements of A. If $A = \{v_1, \dots, v_k\}$, then*

$$\operatorname{span}(A) = \{\lambda_1 v_1 + \cdots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

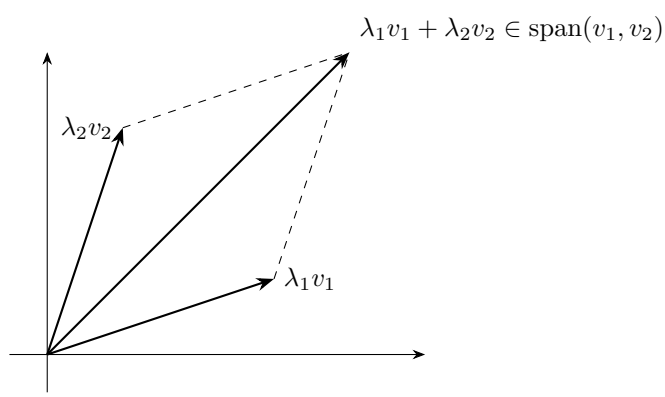
The span describes all points reachable by scaling and adding the given vectors: it can be a line, plane, or higher-dimensional subspace depending on the vectors.

Remark 6. *If $A = \{v\} \subset \mathbb{R}^2$ with $v \neq 0$, then* $\operatorname{span}(v) = \{\lambda v : \lambda \in \mathbb{R}\}$ *is the line through the origin in the direction of v.*

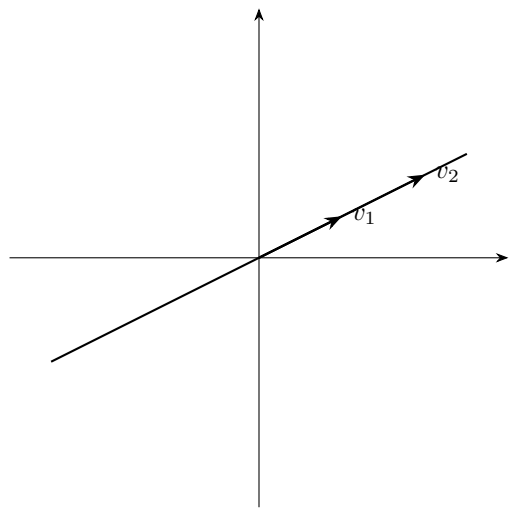
Example. $\operatorname{span}\begin{pmatrix}1\\1\end{pmatrix} = \left\{ t\begin{pmatrix}1\\1\end{pmatrix} \mid t \in \mathbb{R} \right\}$ *is a line in $\mathbb{R}^2$.*

Problem 4 (Span)**.** *Let $v_1 = \begin{pmatrix}3\\1\end{pmatrix}$, $v_2 = \begin{pmatrix}1\\3\end{pmatrix}$. Find* $\operatorname{span}(v_1, v_2)$.

For two linearly independent vectors $v_1, v_2 \in \mathbb{R}^2$, any vector in $\operatorname{span}(v_1, v_2)$ has the form $\lambda_1 v_1 + \lambda_2 v_2$:



If v_1 and v_2 are linearly dependent ($v_1 = \lambda v_2$), then $\operatorname{span}(v_1, v_2)$ is just a line:



Problem 5 (Span 2)**.** *Determine whether the first vector is in the span of the others. If so, write it as a linear combination.*

$$u = (2, 10, 7, 0), \quad u_1 = (3, 10, 7, 0), \quad u_2 = (1, 3, -2, 0), \quad u_3 = (2, 8, 1, 0); \quad u, u_i \in \mathbb{R}^4.$$

3.2.2 Span in $\mathbb{C}^n$

Definition 19 (Span over $\mathbb{C}$)**.**

$$\operatorname{span}_{\mathbb{C}}(v_1, \ldots, v_n) = \{\lambda_1 v_1 + \cdots + \lambda_n v_n : \lambda_i \in \mathbb{C}\}.$$

Example. *Let* $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ i \end{pmatrix} \in \mathbb{C}^2$. *Then* $\lambda_1 v_1 + \lambda_2 v_2 = \begin{pmatrix} \lambda_1 \\ i\lambda_2 \end{pmatrix}$. *Since* λ_1 *and* $i\lambda_2$ *can be any complex number,* $\operatorname{span}_{\mathbb{C}}(v_1, v_2) = \mathbb{C}^2$.

3.2.3 Standard Basis

Definition 20 (Standard basis of $\mathbb{R}^n$)**.** *The standard basis is* $\{e_1, \ldots, e_n\}$, *where* e_i *has a 1 in the* i*-th coordinate and 0 elsewhere:*

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Equivalently, $e_1, \ldots, e_n$ *are the columns of* I_n.

Proposition 1. *Every* $x \in \mathbb{R}^n$ *can be written uniquely as* $x = x_1 e_1 + \cdots + x_n e_n$.[8]

[8] To verify the standard basis is indeed a basis, see Appendix (6.1).

The standard basis also clarifies the difference between real and complex vector spaces — the same vectors generate very different spans depending on the scalar field.

Example (Real vs complex span)**.** *Using e_1, e_2 in $\mathbb{C}^2$: over $\mathbb{C}$, $\operatorname{span}_{\mathbb{C}}(e_1, e_2) = \mathbb{C}^2$ with $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$. Over $\mathbb{R}$, e_1, e_2 cannot produce vectors with imaginary parts. To span $\mathbb{C}^2$ over $\mathbb{R}$, we also need ie_1, ie_2, giving*

$$\dim_{\mathbb{R}} \mathbb{C}^2 = 4.$$

3.3 Examples of Vector Spaces

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3.3.1 Coordinate Spaces

Definition 21 (Coordinate space)**.**

$$k^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in k \right\}, \quad \dim k^n = n.$$

Standard basis: $\{e_1, \ldots, e_n\}$.

3.3.2 Polynomial Spaces

Definition 22 (Polynomial space)**.**

$$P_n(k) = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in k\}, \quad \dim P_n(k) = n + 1.$$

Standard basis: $\{1, x, x^2, \ldots, x^n\}$, with $P_n(k) \cong k^{n+1}$.

Example. $1 + x^2 \in P_2(\mathbb{R})$, $1 + ix^3 \in P_3(\mathbb{C})$.

Remark 7. *The space of all polynomials $P_\infty(k) = \{a_0 + a_1 x + \cdots : a_i \in k\}$ is infinite-dimensional, and the spaces form a nested sequence*

$$P_0(k) \subseteq P_1(k) \subseteq \cdots \subseteq P_\infty(k).$$

3.3.3 Matrix Spaces

Definition 23 (Matrix space)**.**

$$M_n(k) = \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \mid a_{ij} \in k \right\}, \quad \dim M_n(k) = n^2.$$

Standard basis: $\{e_{ij} : 1 \leq i, j \leq n\}$, where e_{ij} has a 1 in the (i, j)-th entry and 0 elsewhere.

Example. *In $M_2(\mathbb{R})$: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1e_{11} + 2e_{12} + 3e_{21} + 4e_{22}$.*

3.3.4 Function Spaces

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Definition 24 (Function space)**.** *Let D be a set and k a field. Define*

$$F(D,k) = \{f : D \to k\},$$

with pointwise operations $(f+g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$. This is a vector space over k.

Finite case. If $D = \{1, \dots, n\}$, define the Kronecker delta functions

$$\delta_i(j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $\{\delta_1, \dots, \delta_n\}$ is a basis of $F(D,k)$ and every f can be written uniquely as $f = \sum_{i=1}^{n} f(i)\,\delta_i$. Hence $\dim F(D,k) = n$.

Infinite case. If D is infinite, $F(D,k)$ has no finite basis; $\{\delta_x : x \in D\}$ is linearly independent but does not span.

Example. *Let $D = \{1,2,3\}$, $k = \mathbb{R}$, and f with $f(1) = 2$, $f(2) = -1$, $f(3) = 3$. The standard basis is $\{\delta_1, \delta_2, \delta_3\}$ and $f = 2\delta_1 - \delta_2 + 3\delta_3$.*

3.4 Subspaces

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3.4.1 Definition and the Subspace Criterion

Definition 25 (Subspace)**.** *A subspace of V is a subset $U \subseteq V$ that is itself a vector space over the same field k.*

Proposition 2 (Subspace criterion)**.** *$U \subseteq V$ is a subspace if and only if:*

1. $0 \in U$,
2. $u, v \in U \Rightarrow u + v \in U$,
3. $u \in U$, $\lambda \in k \Rightarrow \lambda u \in U$.

Remark 8 (Proving a vector space in practice)**.** *To prove that a set U is a vector space over k, you do* ***not*** *always need to verify all 10 axioms. If U is a subset of an already-known vector space V over the same field k, then 7 of the 10 axioms (commutativity, associativity, scalar compatibility, distributivity, etc.) are* inherited automatically *from V. In that case, it suffices to verify only the 3 subspace criteria: $0 \in U$, closure under addition, and closure under scalar multiplication. The full 10-axiom verification is only necessary when constructing a completely new vector space with no ambient space to inherit from.*

Example (Proving a subspace with the 3-criteria)**.** *Let* $V = \mathbb{R}^3$ *and* $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$*. Prove* U *is a subspace.*
Since $U \subset \mathbb{R}^3$*, we only check the 3 criteria.*

1. ***Zero****:* $0 + 0 + 0 = 0$*, so* $(0, 0, 0) \in U$*.* ✓
2. ***Addition****: Let* $u = (x_1, y_1, z_1)$*,* $v = (x_2, y_2, z_2) \in U$*, meaning* $x_1 + y_1 + z_1 = 0$ *and* $x_2 + y_2 + z_2 = 0$*. Then*

   $$(x_1+x_2)+(y_1+y_2)+(z_1+z_2) = (x_1+y_1+z_1)+(x_2+y_2+z_2) = 0+0 = 0,$$

   so $u + v \in U$*.* ✓
3. ***Scalar mult****: Let* $u = (x, y, z) \in U$ *and* $\lambda \in \mathbb{R}$*. Then* $(\lambda x)+(\lambda y)+(\lambda z) = \lambda(x + y + z) = 0$*, so* $\lambda u \in U$*.* ✓

Hence U *is a subspace of* $\mathbb{R}^3$*. Note that* U *is a plane through the origin.*

Problem 6 (Subspace criterion)**.** *Let* $A \in M_n(k)$*. Prove that* $U = \{x \in k^n : Ax = \vec{0}\}$ *is a subspace (the null space or kernel of* A*).*

Problem 7 (Subspace criterion 2)**.** *Let* $A \in M_n(k)$*. Show that* $U = \{Ax \mid x \in k^n\}$ *is a subspace of* k^n *(the image of* A*).*

Problem 8 (Subspace criterion 3)**.** *Let* $V = k$*, viewed as a vector space over* k*. Show that the only subspaces are* $\{0\}$ *and* V *itself.*

Problem 9 (Subspace criterion 4)**.** *Let* $v_1, \ldots, v_n \in V$*. Show that* $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace of* V*.*

3.4.2 Membership in a Span

January 21, 2026

Problem 10 (Span membership)**.** *Let* $v = \begin{pmatrix} 3 \\ 5 \\ -5 \end{pmatrix}$*,* $v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$*,* $v_2 = \begin{pmatrix} 2 \\ 5 \\ 4 \end{pmatrix}$*,* $v_3 = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}$*. Decide whether* $v \in \operatorname{span}(v_1, v_2, v_3)$*.*

Problem 11 (Span membership 2)**.** *Let* $f = 3x^2 + 5x - 5$*,* $f_1 = x^2 + 2x + 1$*,* $f_2 = 2x^2 + 5x + 4$*,* $f_3 = x^2 + 3x + 6$*. Decide whether* $f \in \operatorname{span}(f_1, f_2, f_3)$*.*

3.4.3 Operations on Subspaces

Let W be a vector space and $U, V \subseteq W$ subspaces.

Definition 26 (Intersection)**.**

$$U \cap V = \{w \in W \mid w \in U \text{ and } w \in V\}.$$

The intersection of subspaces is a subspace; the smallest possible is $\{0\}$*.*

Definition 27 (Union)**.**

$$U \cup V = \{w \in W \mid w \in U \text{ or } w \in V\}.$$

The union of subspaces is not *necessarily a subspace.*

Proposition 3. *$U \cup V$ is a subspace of W if and only if $U \subseteq V$ or $V \subseteq U$.*

Proof. If $U \subseteq V$, then $U \cup V = V$, a subspace. Conversely, if neither contains the other, pick $u \in U \setminus V$ and $v \in V \setminus U$. If $U \cup V$ were a subspace, $u+v \in U \cup V$. If $u + v \in U$, then $v = (u + v) - u \in U$, a contradiction. If $u + v \in V$, then $u = (u + v) - v \in V$, also a contradiction. □

Definition 28 (Sum of subspaces)**.**

$$U + V = \{u + v \mid u \in U,\, v \in V\}.$$

Proposition 4. *If U and V are subspaces of W, then $U + V$ is a subspace of W.*

Proof. Apply the subspace criterion: $0 = 0 + 0 \in U + V$; if $w_1 = u_1 + v_1$ and $w_2 = u_2 + v_2$ then $w_1 + w_2 = (u_1 + u_2) + (v_1 + v_2) \in U + V$; for $\lambda \in K$, $\lambda w = \lambda u + \lambda v \in U + V$. □

Proposition 5. *$U + V = \operatorname{span}(U \cup V)$, i.e., $U + V$ is the smallest subspace containing both U and V.*

Proof. ($\subseteq$): Any $u + v \in U + V$ with $u, v \in U \cup V$ lies in $\operatorname{span}(U \cup V)$.
($\supseteq$): Any $w \in \operatorname{span}(U \cup V)$ can be grouped as a sum of a U-part and a V-part, so $w \in U + V$. □

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3.4.4 Direct Sums

Definition 29 (Direct sum)**.** *We say U and V are in direct sum if $U \cap V = \{0\}$, and write $U \oplus V$ for their sum. We write*

$$W = U \oplus V \iff \begin{cases} W = U + V, \\ U \cap V = \{0\}. \end{cases}$$

Proof. $W = U \oplus V$ iff every $w \in W$ decomposes *uniquely* as $w = u + v$. Uniqueness follows since $u_1 - u_2 = v_2 - v_1 \in U \cap V = \{0\}$ forces $u_1 = u_2$. Conversely, if decompositions are unique and $w \in U \cap V$, then $0 = w + (-w) = 0 + 0$ forces $w = 0$. □

Analogy with sets.

Sets	Vector spaces
$A \cap B$	$U \cap V$
$A \cup B$	$U + V$
$A \sqcup B,\ A \cap B = \emptyset$	$U \oplus V,\ U \cap V = \{0\}$

Direct sums in function spaces.

Definition 30 (Support on a subset)**.** *Let $A \subseteq D$. Define $U = \{f \in F(D, \mathbb{R}) : f(x) = 0 \ \forall x \notin A\}$. Then $U \cong F(A, \mathbb{R})$ is a subspace of $F(D, \mathbb{R})$.*

Example. *Let $D = \{1, 2, 3, 4, 5\}$, $A = \{2, 4\}$. Then $U = \{f : f(1) = f(3) = f(5) = 0\}$ is a 2-dimensional subspace, with each $f \in U$ determined by $(f(2), f(4))$.*

4 Basis and Dimension

January 26, 2026

4.1 Finite Dimensional Spaces

Let V be a vector space over a field k. Either $V = \{0\}$, or there exists $v_1 \neq 0$ in V. In the latter case, either $V = \operatorname{span}(v_1)$, or there exists $v_2 \notin \operatorname{span}(v_1)$, and so on. The maximum number of linearly independent vectors is the dimension of V.

Definition 31 (Finite dimensional vector space)**.** *V is finite-dimensional if $\exists$ a finite set $v_1, \ldots, v_n \in V$ with $V = \operatorname{span}(v_1, \ldots, v_n)$. Its dimension is*

$$\dim_k V = \min\{n : V = \operatorname{span}_k(v_1, \ldots, v_n)\}.$$

Example ($\mathbb{C}$ as a real vector space)**.** $\dim_{\mathbb{R}} \mathbb{C} = 2$ *(basis $\{1, i\}$). For $M_n(\mathbb{C})$: $\dim_{\mathbb{C}} M_n(\mathbb{C}) = n^2$ and $\dim_{\mathbb{R}} M_n(\mathbb{C}) = 2n^2$.*

Example (P_∞ is infinite-dimensional)**.** *Since $x^{n+1} \notin P_n = \operatorname{span}\{1, x, \ldots, x^n\}$ for all n, the space $P_\infty(\mathbb{R})$ requires arbitrarily many basis vectors.*

Problem 12 (Infinite dimension of P_∞)**.** *Prove that P_∞ over any field k has infinite dimension.*

4.2 Linear Independence

Definition 32 (Linear independence)**.** *Vectors $v_1, \ldots, v_n$ are linearly independent (LI) if*

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = 0 \implies \lambda_1 = \cdots = \lambda_n = 0.$$

Definition 33 (Linear dependence)**.** *Linearly dependent (LD) is the negation of LI: there exist scalars, not all zero, with $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$.*

Example. $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ *are LI: $xe_1 + ye_2 = \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x = y = 0$.*

Proposition 6. *$v_1, \ldots, v_n$ are LD if and only if one vector lies in the span of the others.*

Proof. ($\Rightarrow$) If $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$ with some $\lambda_k \neq 0$, then $v_k = \frac{-1}{\lambda_k} \sum_{i \neq k} \lambda_i v_i$. ($\Leftarrow$) If $v_k = \sum_{i \neq k} \lambda_i v_i$, rearranging gives a nontrivial relation. $\square$

Remark 9. *If any vector $v_i = 0$, the family is LD.*

Example. *For* $v_1 = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} b_1 \\ b_2 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$, *if* $a_1, b_2, c_3 \neq 0$ *then* v_1, v_2, v_3 *are LI (solved by back-substitution). If* $c_3 = 0$, *all vectors lie in a 2D subspace and are LD.*

January 28, 2026

Problem 13 (Linear independence)**.** *Let* $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$, $v_3 = \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix}$.

Are these LI?

4.3 Basis

4.3.1 Definition and the Main Basis Theorem

Definition 34 (Basis)**.** $(v_1, \ldots, v_n)$ *is a basis of* V *if it spans* V *and is linearly independent.*

Remark 10. *If* V *has a basis of* n *vectors, then* $\dim V = n$.

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Example (Standard basis of $\mathbb{R}^3$)**.** $\{e_1, e_2, e_3\}$ *is a basis of* $\mathbb{R}^3$*: it spans (any* $v = xe_1 + ye_2 + ze_3$*) and is LI (setting to zero gives* $x = y = z = 0$*).*

Lemma 2. *For* $(v_1, \ldots, v_n) \subset \mathbb{R}^k$*: if* $n < k$ *the vectors cannot span* $\mathbb{R}^k$*; if* $n > k$ *they cannot be LI.*

Lemma 3. *If* $u_1, \ldots, u_n$ *span* V *and* $v_1, \ldots, v_m$ *are LI in* V*, then* $n \geq m$.

Theorem 4 (Main Basis Theorem)**.** $(v_1, \ldots, v_n)$ *in* V *are equivalent:*

1. ***Basis****: they span* V *and are LI.*
2. ***Minimal span****: they span* V *and no* $m < n$ *vectors span* V.
3. ***Maximal LI****: they are LI and no* $m > n$ *vectors are LI.*
4. ***Unique coordinates****: every* $v \in V$ *writes uniquely as* $v = \sum_{i=1}^{n} \lambda_i v_i$.

Corollary 4.1 (Coordinate vectors)**.** *For basis* $(v_1, \ldots, v_n)$ *of* V*, every* $v = x_1 v_1 + \cdots + x_n v_n$ *uniquely, and* $[v]_e = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ *is the coordinate vector of* v.

Corollary 4.2. *Any two bases of* V *have the same number of vectors. Hence* $\dim V$ *is well-defined.*

Proof. If $(e_1, \ldots, e_n)$ and $(f_1, \ldots, f_m)$ are both bases, the two-way spanning/independence argument gives $n \geq m$ and $m \geq n$. $\square$

February 2, 2026

4.3.2 Finding a Basis

There are several methods depending on how the space is described.

Method 1: Extracting from a spanning set. Given $v_1, \ldots, v_n \in k^m$ spanning $U = \operatorname{span}(v_1, \ldots, v_n)$:

1. Put the vectors as columns in a matrix A.
2. Row reduce to echelon form.
3. The pivot columns correspond to a basis of U.

Problem 14 (Finding a Basis)**.** *Let V be the subspace of $\mathbb{R}^4$ spanned by $v_1 = \begin{pmatrix}1\\2\\1\\1\end{pmatrix}$, $v_2 = \begin{pmatrix}2\\1\\3\\2\end{pmatrix}$, $v_3 = \begin{pmatrix}1\\2\\0\\1\end{pmatrix}$, $v_4 = \begin{pmatrix}2\\8\\2\\10\end{pmatrix}$. Find a subset forming a basis of V.*

Method 2: Testing linear independence. Solve $a_1v_1 + \cdots + a_nv_n = 0$. If the only solution is the trivial one, the vectors are LI.

Problem 15 (Finding a Basis 2)**.** *Find a basis of the span of $\begin{pmatrix}1 & 2\\1 & 3\end{pmatrix}$, $\begin{pmatrix}2 & 1\\3 & 1\end{pmatrix}$, $\begin{pmatrix}4 & 5\\5 & 7\end{pmatrix}$ in $M_2(\mathbb{R})$.*

Method 3: Constructing from the general form. When a subspace is defined by constraints, write the general element to reveal a basis.

Problem 16 (Finding a Basis 3)**.** *(1) Show $A \in M_2(\mathbb{C})$ is Hermitian iff $A = \begin{pmatrix}x & z\\\bar{z} & y\end{pmatrix}$ for $x, y \in \mathbb{R}$, $z \in \mathbb{C}$.*
(2) Find a basis of the set U of all Hermitian matrices in $M_2(\mathbb{C})$, viewed as a real vector space.

Method 4: Bases of function spaces. If X is a finite set with n elements, $\dim F(X) = n$; check linear independence by evaluating at points.

Problem 17 (Finding a Basis 4)**.** *Let $X = \{-2, -1, 0, 1, 2\}$, $U = \{f \in F(X) : f(-x) = f(x)\ \forall x\}$. With $g_1(x) = 1$, $g_2(x) = \frac{1}{4}x^2(1 - x^2)$, $g_3(x) = \frac{1}{3}x^2(4 - x^2)$, prove $S = (g_1, g_2, g_3)$ is a basis of U.*

4.3.3 Proving a Set Forms a Basis

Method 1 (Dimension test)**.** *$(v_1, \ldots, v_n)$ is a basis of V iff $\dim(\operatorname{span}(v_1, \ldots, v_n)) = n$.*

Proposition 7 (Basis criterion)**.** *If $\dim V = n$, then for $v_1, \dots, v_n \in V$, the following are equivalent: (1) basis, (2) spans V, (3) LI.*

Proof. A spanning set of $n = \dim V$ vectors is minimal, hence LI. An independent set of n vectors is maximal, hence spans V. □

Principle 1 (Transfer/Exchange principle)**.** *If $u_1, \dots, u_n$ span V and $v_1, \dots, v_n$ are LI, then both lists are bases of V.*

Example. *In P_n: $(1, x, \dots, x^n)$ spans and $((x-a)^0, \dots, (x-a)^n)$ is LI, so both are bases of P_n.*

February 4–6, 2026

4.4 Coordinates and Change of Basis

4.4.1 Coordinate Vectors

If $(e_1, \dots, e_n)$ is a basis of V, every $v \in V$ writes uniquely as $v = \lambda_1 e_1 + \dots + \lambda_n e_n$, and

$$[v]_e = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}.$$

Example (Finding a coordinate vector)**.** *Find the coordinate vector of $v = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}$ in the basis $f = (f_1, f_2, f_3)$ where $f_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $f_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.*
We need x, y, z such that $xf_1 + yf_2 + zf_3 = v$. Writing this as a system:

$$\begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}.$$

Row reduce the augmented matrix:

$$\begin{pmatrix} 1 & 1 & 0 & 5 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2+R_1} \begin{pmatrix} 1 & 1 & 0 & 5 \\ 0 & 2 & 1 & 8 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2-R_3} \begin{pmatrix} 1 & 1 & 0 & 5 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$$

Back-substitution: $z = 4$, $y = 2$, $x = 5 - 2 = 3$. Therefore $[v]_f = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$.

Problem 18 (Coordinate vectors)**.** *Let $X = \{-2, -1, 0, 1, 2\}$, $U = \{f \in F(X) : f(-x) = f(x)\}$, with basis $S = (g_1, g_2, g_3)$ as above. Find $[g]_S$ for g with $g(\pm 2) = 3$, $g(\pm 1) = g(0) = 2$.*

Problem 19 (Coordinate vectors 2)**.** *Let $D = \{1, 2, 3\}$. Find the coordinates of $f(x) = 2x^2 - 3$ in the standard basis of $F(D)$.*

4.4.2 Direct Sum Decompositions and Dimension Formulas

Theorem 5 (Basis concatenation)**.** *If $V = U \oplus U'$ with bases $(e_1, \ldots, e_n)$ and $(f_1, \ldots, f_m)$ respectively, then $(e_1, \ldots, e_n, f_1, \ldots, f_m)$ is a basis of V and $\dim(U \oplus U') = \dim U + \dim U'$.*

Proof. LI: $\sum a_i e_i + \sum b_j f_j = 0$ gives $w = \sum a_i e_i = -\sum b_j f_j \in U \cap U' = \{0\}$, forcing all $a_i = b_j = 0$. Span: every $v \in U + U'$ writes $v = u + u'$. □

Theorem 6 (Dimension of sum)**.**

$$\dim(U + U') = \dim U + \dim U' - \dim(U \cap U').$$

Corollary 6.1. *If $U \subseteq V$, then $\dim U \leq \dim V$.*

Corollary 6.2. *If $U \subseteq V$, then U admits a complement: $\exists U' \subseteq V$ with $V = U \oplus U'$ and $\dim V = \dim U + \dim U'$.*

February 11, 2026

4.5 Basis Extension and Extraction

Theorem 7 (Basis extension)**.** *Let V be finite-dimensional and $v_1, \ldots, v_m$ LI in V. Then they can be extended to a basis of V.*

Proof. Set $U = \operatorname{span}(v_1, \ldots, v_m)$. If $U = V$, done. Otherwise pick $v_{m+1} \in V \backslash U$; this gives a larger LI set. Repeat until the span equals V, which happens after at most $\dim V - m$ steps. □

Theorem 8 (Basis extraction)**.** *If $v_1, \ldots, v_n$ span V, one can extract a basis from this list.*

5 Linear Transformations

5.1 Functions: Basic Definitions

Let U and V be sets. A transformation $T : U \to V$ assigns to each $u \in U$ a unique element $T(u) \in V$. The set U is the domain, V the codomain, and $\operatorname{Im}(T) = \{T(u) : u \in U\} \subseteq V$ is the image.

Definition 35 (Injective, surjective, bijective)**.**

- *T is **surjective** if $\operatorname{Im}(T) = V$.*
- *T is **injective** if $T(u) = T(u') \implies u = u'$.*
- *T is **bijective** if it is both injective and surjective.*

Example. *$f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$ is bijective. $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$ is neither injective nor surjective.*

Example. *$z \mapsto z^2$ on $\mathbb{C}$: not injective since $(-1)^2 = 1^2$, but surjective since every $w \in \mathbb{C}$ has a square root.*

5.1.1 Composition and Inverses

Given $T : U \to V$ and $S : V \to W$, the composition $R = S \circ T : U \to W$ is defined by $R(u) = S(T(u))$.

Example. *$g = \sin \circ (x^2) : \mathbb{R} \xrightarrow{x^2} \mathbb{R} \xrightarrow{\sin} \mathbb{R}$, so $g(x) = \sin(x^2)$.*

Proposition 8. *$T : U \to V$ is bijective $\iff$ T has an inverse $S : V \to U$ with $ST = \mathrm{id}_U$ and $TS = \mathrm{id}_V$.*

5.2 Linear Maps

Definition 36 (Linear transformation)**.** *Let U, V be vector spaces over k. A map $T : U \to V$ is **linear** if:*

1. *Additive: $T(u + u') = T(u) + T(u')$,*
2. *Scalar: $T(\lambda u) = \lambda T(u)$ for all $\lambda \in k$.*

5.2.1 Main Examples of Linear Maps

Matrix transformations. The primary example is $T_A : \mathbb{R}^n \to \mathbb{R}^n$, $T_A(x) = Ax$ for a fixed $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \implies T_A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}.$$

Scaling.

Definition 37 (Scaling transformation)**.** *$T : U \to U$ is called scaling if there exists a basis $(u_1, \ldots, u_n)$ and scalars $\lambda_1, \ldots, \lambda_n \in k$ such that $T(u_i) = \lambda_i u_i$ for all i.*

The derivative.

$$D : P_n \to P_n, \quad f \mapsto f', \quad D(x^k) = kx^{k-1}.$$

D is linear: $(f + g)' = f' + g'$ and $(\lambda f)' = \lambda f'$. D is not injective ($D(1) = 0$) and not surjective ($x^n \notin \mathrm{Im}(D)$ since every polynomial in $\mathrm{Im}(D)$ has degree $\leq n - 1$).

The indefinite integral.

$$\int : P_\infty \to P_\infty \quad \text{[linear, injective, not surjective]}.$$

Injective: if $\int f = 0$ then $f = 0$ (degree of $\int f$ exceeds degree of f). Not surjective: no $f \in P_\infty$ satisfies $\int f = 1$.

Projection. Suppose $V = U \oplus U'$. Every $v \in V$ writes uniquely as $v = u + u'$. Define $P(v) := u$.

Proposition 9. *The projection $P : V \to U$ along U' is linear.*

Proof. Write $v_1 = u_1 + u_1'$ and $v_2 = u_2 + u_2'$. Then $v_1 + v_2 = (u_1 + u_2) + (u_1' + u_2')$, so $P(v_1 + v_2) = u_1 + u_2 = P(v_1) + P(v_2)$. Scalar case is analogous. □

5.2.2 Defining a Linear Map by its Action on a Basis

Proposition 10. *Let $(u_1, \ldots, u_n)$ be a basis of U and $v_1, \ldots, v_n$ arbitrary vectors in V. There exists a unique linear $T : U \to V$ with $T(u_i) = v_i$.*

Proof. Existence: for any $u = \lambda_1 u_1 + \cdots + \lambda_n u_n$, define $T(u) = \lambda_1 v_1 + \cdots + \lambda_n v_n$. Linearity follows from uniqueness of coordinates.
Uniqueness: any linear T with $T(u_i) = v_i$ is forced on all u by linearity. □

Proposition 11. *If S, T are linear, then $S \circ T$ is linear.*

5.3 Kernel and Image

February 18, 2026

Definition 38 (Kernel and image)**.**

$$\ker(T) = \{u \in U : T(u) = 0\}, \qquad \operatorname{Im}(T) = \{T(u) : u \in U\} \subseteq V.$$

Proposition 12. $\ker(T) \subseteq U$ *and* $\operatorname{Im}(T) \subseteq V$ *are subspaces.*

Proposition 13. *T is injective* $\iff$ $\ker(T) = \{0\}$.

Proof. ($\Leftarrow$) If $\ker(T) = \{0\}$ and $T(u) = T(u')$, then $T(u - u') = 0$, so $u - u' = 0$.
($\Rightarrow$) Contrapositive: if $\exists u \neq 0$ with $T(u) = 0$, then $T(u) = T(0)$ with $u \neq 0$. □

Remark 11. *The kernel is only useful for linear T. For example,* $\ker(\sin) = \{k\pi : k \in \mathbb{Z}\}$ *is not a subspace.*

Proposition 14. *If $(e_1, \ldots, e_n)$ is a basis of U, then* $\operatorname{Im}(T) = \operatorname{span}(T(e_1), \ldots, T(e_n))$.

Proof. Any $v = T(u) = T(\lambda_1 e_1 + \cdots + \lambda_n e_n) = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n) \in \operatorname{span}(T(e_i))$. □

5.3.1 Examples of Kernel and Image

Example. *For the projection $P : V \to U$ along U':* $\operatorname{Im}(P) = U$ *and* $\ker(P) = U'$.

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $T_A : \mathbb{R}^2 \to \mathbb{R}^2$. *Solving* $\begin{cases} x + 2y = 0 \\ 3x + 4y = 0 \end{cases}$ *gives* $x = y = 0$, *so* $\ker(T_A) = \{0\}$ *and T_A is injective.*

Example. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. *Row reduction gives* $\ker(T_A) = \operatorname{span}\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.

February 20, 2026

5.4 Matrix of a Linear Transformation

Definition 39 (Matrix of T in basis e)**.** *Let $e = (e_1, \ldots, e_n)$ be a basis of U. The matrix of $T : U \to U$ in basis e is*

$$[T]_e = \big([T(e_1)]_e, \ldots, [T(e_n)]_e\big).$$

The columns are the coordinate vectors of the images of the basis vectors.

Example. *For $T = T_A$ on $\mathbb{R}^2$ with the standard basis, $[T]_e = A$.*

Example (Derivative matrix for P_2)**.** *D is the derivative operator: $D(f) = f'$. It is linear since $(f+g)' = f' + g'$ and $(\lambda f)' = \lambda f'$. The goal is to represent it as a matrix so that differentiating a polynomial becomes a matrix-vector multiplication.*

Working in P_2 with basis $e = (1, x, x^2)$, every polynomial $a + bx + cx^2$ has coordinate vector $(a, b, c)^T$. To build $[D]_e$, we apply D to each basis vector and write the result's coordinates as a column:

$$D(1) = 0 \to \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad D(x) = 1 \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad D(x^2) = 2x \to \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix},$$

so $[D]_e = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$. *To verify: $f = 3 + 5x - 2x^2$ has $f' = 5 - 4x$, and indeed*

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 0 \end{pmatrix} \longleftrightarrow 5 - 4x. \checkmark$$

Example (Matrix of T in a non-standard basis)**.** *Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix}$, and let $f = (f_1, f_2)$ with $f_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Find $[T]_f$.*

The recipe: apply T to each basis vector, then express the result in terms of the basis f.

Column 1. $T(f_1) = T\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, *so* $[T(f_1)]_f = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$.

Column 2. $T(f_2) = T\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 0\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, *so* $[T(f_2)]_f = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

Therefore $[T]_f = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$, *which is diagonal! This means f_1 and f_2 are eigenvectors of T with eigenvalues 4 and 2. Choosing the right basis makes the matrix much simpler.*

February 23, 2026

5.5 Isomorphisms

Definition 40 (Isomorphism)**.** *$T : U \to V$ is an isomorphism if it is linear and bijective.*

To check: verify T is linear, $\ker(T) = \{0\}$, and $\mathrm{Im}(T) = V$.

Theorem 9. *For $T : V \to V$ with matrix A in basis e, the following are equivalent:*

1. *T is an isomorphism.*
2. *A is invertible.*
3. $\det(A) \neq 0$.

Theorem 10. *$U \cong V \iff \dim U = \dim V$. In particular, every n-dimensional space is isomorphic to k^n.*

Example. *$P_8 \cong \mathbb{R}^9 \cong M_3(\mathbb{R})$ since all have dimension 9.*

5.5.1 The Canonical Isomorphism

Definition 41 (Coordinate map)**.** *Given vectors $v_1, \ldots, v_n \in V$, define $\Phi : k^n \to V$ by*

$$\Phi \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \lambda_1 v_1 + \cdots + \lambda_n v_n.$$

Then $\mathrm{Im}(\Phi) = \mathrm{span}(v_1, \ldots, v_n)$.

Theorem 11. *Φ is an isomorphism $\iff$ $(v_1, \ldots, v_n)$ is a basis for V. In this case, $[v]_e = \Phi^{-1}(v)$.*

Example (Using isomorphism to prove LI)**.** *To show x and x^2 are LI in P_2, use $\Phi : P_2 \to \mathbb{R}^3$, $a + bx + cx^2 \mapsto (a, b, c)^T$, and verify $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are LI in $\mathbb{R}^3$.*

Example (Exam-style)**.** *Show $\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ are LI using the isomorphism $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$.*

February 27 – March 13, 2026

5.6 Change of Basis

5.6.1 Change of Basis Matrix

Let $e = (e_1, \ldots, e_n)$ and $f = (f_1, \ldots, f_n)$ be two bases of V. Define:

- Q: change of basis matrix from f-coordinates to e-coordinates. Its columns are $[f_1]_e, \ldots, [f_n]_e$.
- $P = Q^{-1}$: change from e-coordinates to f-coordinates.

Example. *$f_1 = (1,0)^T$, $f_2 = (1,1)^T$ gives $Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $P = Q^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.*

5.6.2 Change of Basis Formula

Theorem 12 (Change of basis)**.** *If $A = [T]_e$ and $B = [T]_f$, then*

$$B = PAQ, \quad P = Q^{-1}.$$

Example (Full change of basis computation)**.** *Let e be the standard basis of $\mathbb{R}^2$, $f_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $f_2 = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, and $T : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x - 7y \\ 4x - 3y \end{pmatrix}$, so $A = [T]_e = \begin{pmatrix} 2 & -7 \\ 4 & -3 \end{pmatrix}$. Find $[T]_f$.*
Step 1: Build Q. *The columns of Q are the f-basis vectors in e-coordinates (which are just themselves since e is the standard basis):*

$$Q = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}.$$

Step 2: Find $P = Q^{-1}$. *For a 2×2 matrix, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$:*

$$P = Q^{-1} = \frac{1}{5-6} \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}.$$

Step 3: Compute $B = PAQ$.

$$AQ = \begin{pmatrix} 2 & -7 \\ 4 & -3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 2-21 & 4-35 \\ 4-9 & 8-15 \end{pmatrix} = \begin{pmatrix} -19 & -31 \\ -5 & -7 \end{pmatrix}.$$

$$B = P(AQ) = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} -19 & -31 \\ -5 & -7 \end{pmatrix} = \begin{pmatrix} 95-10 & 155-14 \\ -57+5 & -93+7 \end{pmatrix} = \begin{pmatrix} 85 & 141 \\ -52 & -86 \end{pmatrix}.$$

So $[T]_f = \begin{pmatrix} 85 & 141 \\ -52 & -86 \end{pmatrix}$.

March 13–16, 2026

5.7 Eigenvalues and Diagonalization

5.7.1 Eigenvalues and Eigenvectors

Definition 42 (Eigenvalue and eigenvector)**.** *$\lambda \in k$ is an eigenvalue of T if $\exists\, v \neq 0$ with $Tv = \lambda v$; v is an eigenvector. The eigenspace is*

$$E_\lambda = \{v \in V : Tv = \lambda v\}.$$

Proposition 15. *E_λ is a subspace of V.*

Proposition 16. *If $\lambda_1 \neq \lambda_2$, then $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.*

Proof. If $v \in E_{\lambda_1} \cap E_{\lambda_2}$, then $\lambda_1 v = Tv = \lambda_2 v$, so $(\lambda_1 - \lambda_2)v = 0$. Since $\lambda_1 \neq \lambda_2$, we get $v = 0$. □

5.7.2 The Characteristic Polynomial

Definition 43 (Characteristic polynomial)**.**

$$\chi(\lambda) = \det(\lambda \operatorname{Id} - A).$$

Theorem 13. *λ is an eigenvalue of $A \iff \lambda$ is a root of χ.*

5.7.3 Diagonalization

Definition 44 (Diagonalizable)**.** *T (or its matrix A) is diagonalizable if $\exists$ a basis $e = (e_1, \ldots, e_n)$ and scalars $\lambda_1, \ldots, \lambda_n$ with $T(e_i) = \lambda_i e_i$ (i.e., T is a scaling map).*

Proposition 17. *$T_A : \mathbb{R}^n \to \mathbb{R}^n$ is scaling $\iff$ A is diagonalizable.*

How to diagonalize A.

1. Compute $\chi(\lambda) = \det(\lambda \operatorname{Id} - A)$ and find its roots (the eigenvalues).
2. For each eigenvalue λ_i, find $\ker(\lambda_i \operatorname{Id} - A)$ to get eigenvectors.
3. If there are n linearly independent eigenvectors, write them as columns of P; then $P^{-1}AP$ is diagonal.

Example (Full diagonalization of a 2×2 matrix)**.** *Diagonalize $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ over $\mathbb{R}$.*
Step 1: Characteristic polynomial.

$$\chi(\lambda) = \det \begin{pmatrix} \lambda - 4 & -1 \\ -2 & \lambda - 3 \end{pmatrix} = (\lambda - 4)(\lambda - 3) - (-1)(-2) = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2).$$

Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.
Step 2: Eigenvectors.
For $\lambda_1 = 5$*: solve* $(5I - A)v = 0$*:*

$$\begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x = y, \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 2$*: solve* $(2I - A)v = 0$*:*

$$\begin{pmatrix} -2 & -1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies 2x + y = 0, \quad v_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Step 3: Form P ***and verify.***

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad P^{-1} = \frac{1}{-3} \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{pmatrix}.$$

$$P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

The diagonal entries are exactly the eigenvalues.

6 Appendix

6.1 Proof: Standard Basis is a Basis

Claim 1. $e_1, \ldots, e_n$ span $\mathbb{R}^n$.

Proof. $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 e_1 + \cdots + x_n e_n.$ □

Claim 2. $e_1, \ldots, e_n$ are linearly independent.

Proof. $\lambda_1 e_1 + \cdots + \lambda_n e_n = 0 \implies \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = 0 \implies \lambda_i = 0.$ □

6.2 Algebraic Structures: Rings, Fields, and Vector Spaces

- **Ring**: scalars come from inside the set itself; no scalar-vector distinction.
- **Vector space**: scalars come from an external field F; multiplication between vectors is not defined, only scalar multiplication.

7 Solutions

Solution 1 (Hermitian Matrix)**.** $(AB)^* = (\overline{AB})^T = (\overline{A}\,\overline{B})^T = \overline{B}^T\overline{A}^T = B^*A^*$.

Solution 2 (Invertibility)**.** *Suppose B and B' are both inverses of A. Then $B = BI = B(AB') = (BA)B' = IB' = B'$.*

Solution 3 (Invertibility 2)**.** *Suppose the matrix is invertible. Then* $\det\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} = 0\cdot 0 - 1\cdot 0 = 0$*, but an invertible matrix must have nonzero determinant. Contradiction.*

Solution 4 (Span)**.** $\operatorname{span}(v_1, v_2) = \left\{\begin{pmatrix}3x+y\\ x+3y\end{pmatrix} : x, y \in \mathbb{R}\right\}$.

Solution 5 (Span 2)**.** *Setting up $u = au_1 + bu_2 + cu_3$ gives the augmented matrix* $\left(\begin{array}{ccc|c}3 & 1 & 2 & 2\\ 10 & 3 & 8 & 10\\ 7 & -2 & 1 & 7\end{array}\right)$*. Gaussian elimination gives $a = \frac{2}{21}$, $b = -\frac{46}{21}$, $c = \frac{41}{21}$, so $u \in \operatorname{span}(u_1, u_2, u_3)$.*

Solution 6 (Subspace criterion)**.** *$U = \{x \in k^n : Ax = 0\}$. Check: $A0 = 0$ so $0 \in U$; if $Ax = Ay = 0$ then $A(x+y) = 0$; if $Ax = 0$ then $A(\lambda x) = 0$. So U is a subspace (the null space of A).*

Solution 7 (Subspace criterion 2)**.** *$U = \{Ax : x \in k^n\}$. Check: $A0 = 0$; $Ax_1 + Ax_2 = A(x_1 + x_2)$; $\lambda Ax = A(\lambda x)$. So U is a subspace (the image of A).*

Solution 8 (Subspace criterion 3)**.** *If $V \neq \{0\}$, pick $v \in V$, $v \neq 0$. For any $x \in k$, write $x = \lambda v$ for $\lambda = x/v \in k$. By closure under scalar multiplication, $x \in V$. Hence $V = k$.*

Solution 9 (Subspace criterion 4)**.** $0 = 0v_1 + \cdots + 0v_n \in \operatorname{span}$*; sums and scalar multiples of linear combinations are linear combinations. Hence* $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace.*

Solution 10 (Span membership)**.** *Solve $xv_1 + yv_2 + zv_3 = v$: the system gives $x = 3$, $y = 1$, $z = -2$, so $v = 3v_1 + v_2 - 2v_3 \in \operatorname{span}(v_1, v_2, v_3)$.*

Solution 11 (Span membership 2)**.** *Comparing coefficients gives the same system as above: $t = 3$, $y = 1$, $z = -2$, so $f = 3f_1 + f_2 - 2f_3$.*

Solution 12 (Infinite dimension of P_∞)**.** *If $\dim P_\infty < \infty$, some $f_1, \ldots, f_n$ span P_∞. Let $d = \max\deg(f_i)$. Then every polynomial in* $\operatorname{span}(f_1, \ldots, f_n)$ *has degree $\leq d$, but $x^{d+1} \notin \operatorname{span}$. Contradiction.*

Solution 13 (Linear independence)**.** $3v_1 + 5v_2 - 2v_3 = 0$ *(with $x = 3, y = 5, z = -2$), so the vectors are not LI.*

Solution 14 (Coordinate vectors)**.** *Evaluating at $x = 0, 1, 2$: $\lambda_1 = 2$, $\lambda_3 = 0$, $\lambda_2 = -\frac{1}{3}$. So $[g]_S = \begin{pmatrix}2\\ -1/3\\ 0\end{pmatrix}$.*

Solution 15 (Coordinate vectors 2)**.** *Evaluate:* $f(1) = -1$, $f(2) = 5$, $f(3) = 15$. *So* $[f]_e = \begin{pmatrix} -1 \\ 5 \\ 15 \end{pmatrix}$.

Solution 16 (Finding a Basis)**.** *Row reducing the matrix with* v_1, v_2, v_3, v_4 *as columns yields four pivots; all four vectors are LI and form a basis of* V.

Solution 17 (Finding a Basis 2)**.** *Solving* $a_1M_1 + a_2M_2 + a_3M_3 = 0$ *gives* $a_3 = 1$, $a_2 = -1$, $a_1 = -2$, *so* $M_3 = 2M_1 + M_2$. *Basis:* $\left\{ \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \right\}$.

Solution 18 (Finding a Basis 3)**.** *Decomposing* $A = \begin{pmatrix} x & a+bi \\ a-bi & y \end{pmatrix}$ *gives basis* $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \right\}$.

Solution 19 (Finding a Basis 4)**.** $\dim U = 3$ *(three free values:* $f(0), f(1), f(2)$*). Evaluating at* $x = 0, 1, 2$ *forces* $c_1 = c_3 = c_2 = 0$, *so* S *is LI. Since* $|S| = \dim U = 3$, S *is a basis.*

8 Useful Links

Proofs: https://www.youtube.com/watch?v=V5tUc-J124s